

CHILLIWORKS GAME ENGINE PARTICLE OPTIMIZATION

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This is the short summary of why this thesis is awesome and *YOU* really want to read it!

ACKNOWLEDGMENTS

Thanks guys ...

TABLE OF CONTENTS

COPYRIGHT	ii
ABSTRACT	iii
ACKNOWLEDGMENTS	iv
CODE LISTINGS	vii
LIST OF FIGURES	viii
LIST OF TABLES	ix
CHAPTER 1 INTRODUCTION	1
1.1 Thesis Organization	1
CHAPTER 2 BACKGROUND	2
2.1 Particle Systems	2
2.1.1 Overview	2
2.1.2 Particle Model	3
2.1.2.1 Life Cycle	3
2.1.2.2 Particle Attributes	4
2.1.2.3 Hierarchy	5
2.2 The ChilliSource Game Engine	6
2.2.1 Overview	6
2.2.2 Architecture	6

2.2.3	Life Cycle	6
2.3	ChilliSource Particle Effect Components	6
2.3.1	Overview	6
2.3.2	Usage	6
2.3.3	Architecture	6
2.3.4	Attributes	6
2.3.5	Life Cycle	6
2.3.5.1	Overview	6
2.3.5.2	Update Event	6
2.3.5.3	Render Event	6
CHAPTER 3 THE CHILLISOURCE AUTOMATED PONG GAME		7
CHAPTER 4 CHILLISOURCE PARTICLE EFFECT COMPO-		
NENT CASE STUDIES		8
CHAPTER 5 CHILLISOURCE PARTICLE EFFECT COMPO-		
NENT OBSERVATIONS		9
BIBLIOGRAPHY		10

CODE LISTINGS

LIST OF FIGURES

Figure 2.1	Particle effect created by William T. Reeves	2
Figure 2.2	Cubes emitting particles	3

LIST OF TABLES

CHAPTER 1 INTRODUCTION

1.1 Thesis Organization

The rest of this thesis is organized as follows:

- **Chapter 2** Review of particle systems and the ChilliSource game engine
- **Chapter 3** Overview of the automated pong game that was created to facilitate case studies
- **Chapter 4** Discussion of the two case studies performed to explore contention in large data structures
- **Chapter 5** Discussion of three interesting aspects of the ChilliSource particle effect component

CHAPTER 2 BACKGROUND

2.1 Particle Systems

2.1.1 Overview

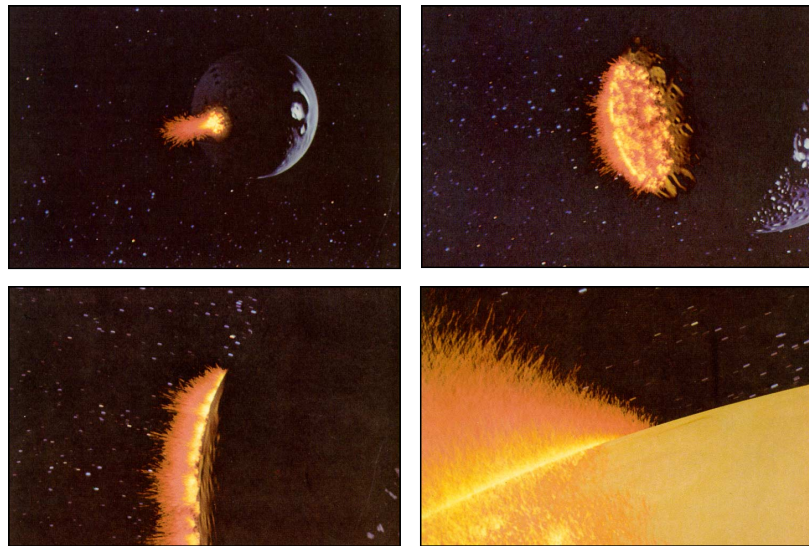


Figure 2.1: The particle effect created by William T. Reeves from the film *Star Trek II: The Wrath of Khan*

Visual representation of "fuzzy objects", or objects that do not have well-defined surfaces, in computer graphics is generally considered a difficult task due to the non-deterministic changes in the fuzzy object's shape and form. When working on the film *Star Trek II: The Wrath of Khan*, a researcher at Lucasfilm Ltd., William Reeves, developed what he called a **particle system**, which is "... a collection of many minute

particles that together represent a fuzzy object. Over a period of time, particles are generated into a system, move and change from within the system, and die from the system.” [1] Although the concept of modeling these objects as a collection of particles was not novel, Reeves’ basic framework for particle systems is still widely used today to create complex visual effects in video games and films.

Fuzzy objects like sparks, clouds, stars, water, and crowds of people can be easily translated and modeled as a particle system, but particle systems can also be used to model objects that contain numerous strands like hair and grass. These objects can be represented by rendering every strand as the entire lifetime of a particle in the system and is then treated as a single strand. The particles in a system that have a distributed lifetime (e.g. sparks) are thought of as **animated** and particles that are rendered all at once (e.g. hair) are thought of as **static**.

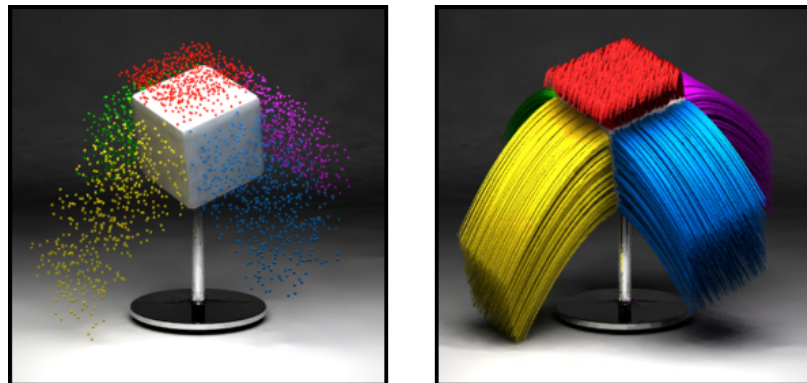


Figure 2.2: Left: A cube emitting a stream of 1000 animated particles from 5 of its 6 faces. Right: A cube emitting a stream of 1000 static particles, or strands, from 5 of its 6 faces

2.1.2 Particle Model

2.1.2.1 Life Cycle

A particle system has 4 main events that defines its life cycle:

1. Emission (or Generation)
2. Update
3. Render
4. Death (or Extinction)

The **Emission** event is the initial process in which particles are generated with their initial values. Often a "generation shape" is used during the emission phase that defines the two or three dimensional space that the particles can originate, or emit, from. Emission can be increased or decreased in frequency with the *emission rate*, and the number of particles that enter the particle system each emission can also be changed. After emission, particles in the particle system undergo the **Update** and **Render** events. These events are repeated in perpetuum until the particle dies, or undergoes the extinction phase. The update phase involves iterating over all particles throughout the system and updating their attributes (e.g. position, velocity, etc.). The render phase, on the other hand, is the phase in which all of the particles are actually drawn to the screen.

2.1.2.2 Particle Attributes

Each individual particle in a particle system will have attributes that change over time. Although every particle system is different, the following is what Reeves considers to be the core attributes of a particle:

- Position
- Velocity
- Size
- Color
- Transparency
- Shape
- Lifetime

A particle has a position in two or three dimensional space, and its initial position is determined by the emission (or, as Reeves called it, generation) process. Other attributes that may also change over time include its speed, direction, size, color, and transparency. Its lifetime, however, is pre-determined at emission, and it is the attribute that dictates how long the particle "lives", or remains on the screen. In order to emulate realism and decrease uniformity, all of these values can vary across the particles in the system. For example, the particles in a given particle system could have a varying lifetime of 5 to 10 seconds so that the particles that shared an emission would not all die simultaneously.

2.1.2.3 Hierarchy

It is often convenient to design a particle system such that it is made up of a hierarchy of child particle systems. This hierarchy "can be used to exert global control on a complicated fuzzy object that is composed of many particle systems." [1] In other words, a fuzzy object could easily be transformed in overall shape, size, color, etc. by sweeping through the descendant particle systems. It also allows the grouping together of particle systems that have different types of particles. For example, a campfire fuzzy object would need smoke, sparks, and the flame itself. All of these components would have distinct particle definitions and would be separate particle systems. A particle system hierarchy, in this case, would allow the smoke, sparks, and flame particle systems to be bundled together as a single fuzzy object.

2.2 The ChilliSource Game Engine

2.2.1 Overview

2.2.2 Architecture

2.2.3 Life Cycle

2.3 ChilliSource Particle Effect Components

2.3.1 Overview

2.3.2 Usage

2.3.3 Architecture

2.3.4 Attributes

2.3.5 Life Cycle

2.3.5.1 Overview

2.3.5.2 Update Event

2.3.5.3 Render Event

CHAPTER 3 THE CHILLISOURCE AUTOMATED PONG GAME

CHAPTER 4 CHILLISOURCE PARTICLE EFFECT COMPONENT CASE STUDIES

CHAPTER 5 CHILLISOURCE PARTICLE EFFECT COMPONENT OBSERVATIONS

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